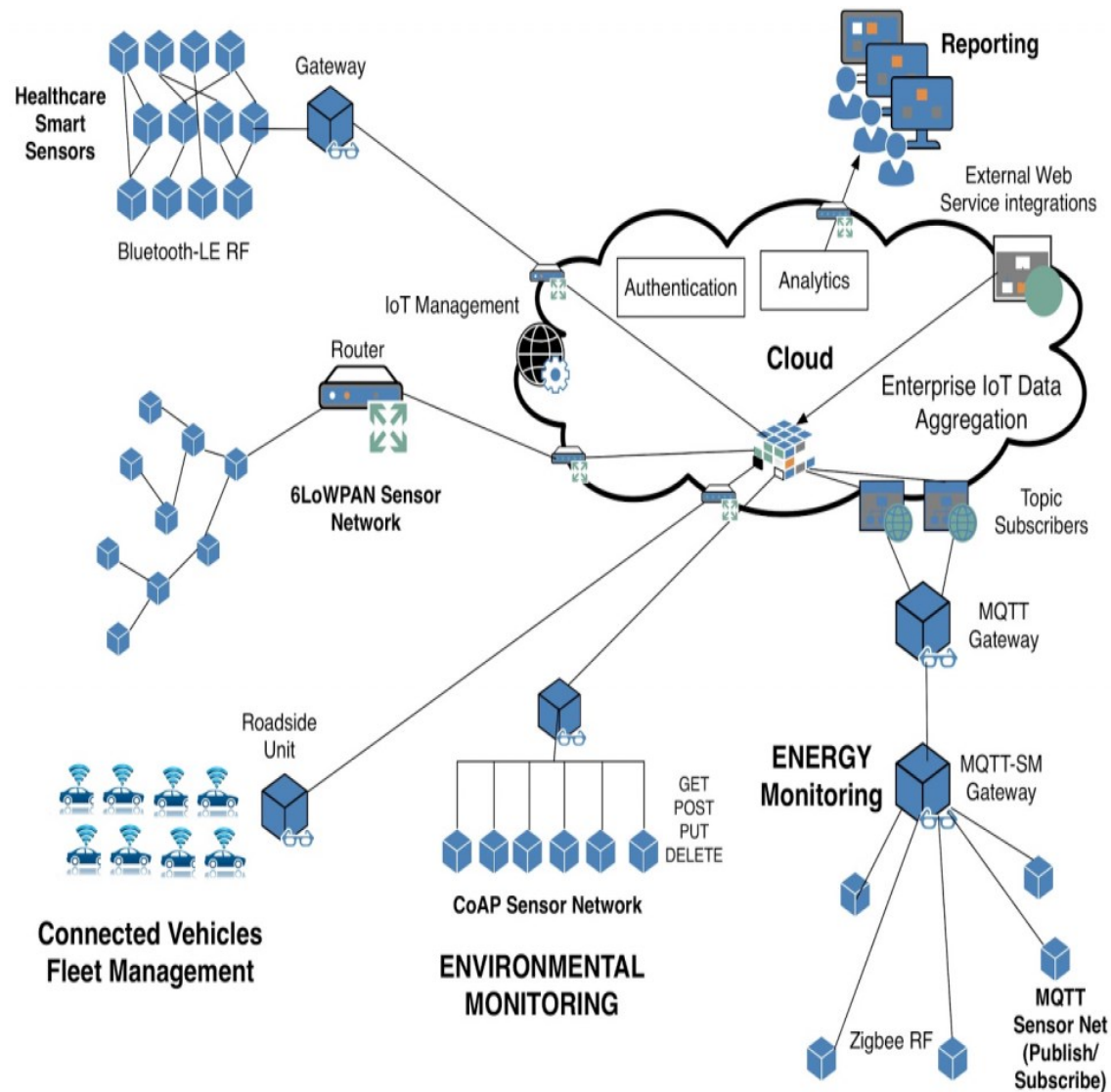


- A human driving home wants to watch a football game, eat pizza and drink cold beer. The smart car coordinates with the smart fridge to keep the beer cold, with the smart oven to get the pizza cooked and keep it warm, with the smart TV set to switch to the right channel, and potentially with the smart video recorder to record the football game if there is too much traffic. At the same time, there is also a coordination through satellites and smart navigation systems in order to calculate the best route for the smart car, estimate its arrival time, etc., and this information is directly or indirectly shared with the smart home devices.
 - Car / satellite / navigation system / fridge / oven / TV set / video recorder

- A forest fire may occur near a highway. Such a fire can be detected by the space segment of IoT, which can, potentially, also use any on-location IoT devices, which may, for example, be placed on nearby fields for agricultural reasons to confirm the existence of the fire and its extent. Then, the IoT devices on the space satellites that have detected the fire, as well as the on-location IoT devices, can forward their sensor data to the nearby highway infrastructure as well as to oncoming traffic for the appropriate traffic signs to be displayed on any nearby configurable LED indicator boards and for the appropriate actions to be taken by the vehicles, either autonomously or through driver notifications. Additionally, smart vehicles can pass this information to nearby vehicles, so that traffic can be diverted in such a way as to avoid any danger. Finally, all IoT devices involved can potentially also notify the closest fire department, which may be employing supervised and/or autonomous IoT devices and vehicles that can be directed to the spot based on the information exchanged on the IoT, either without any human intervention being needed or with minimal supervision mechanisms being required. In this example case, through the use of the IoT and its related technology, including sensors, actuators, and other electronics, a forest fire that may have been started by human error could potentially be put out completely autonomously. At the same time, the same system can also ensure that any nearby humans are kept out of harm as much as possible.
 - Satellite / on-site (agricultural?) IoT / highway traffic coordination infrastructure / cars / fire truck / smaller firefighting systems / other emergency vehicles / firefighting and coordination UAVs / cellphones

IoT Applications: Another Example



Russel & Van Duren: "Practical Internet of Things Security", 2nd edition, Packt Publishing, 2019

- Internet of lights
 - “An Internet of Lights is formed when all the smart lights in a space are connected by digital networks and are able to communicate with each other, communicate with a server or gateway, and communicate with sensors and controls in proximity of the lights.”

(https://www.photonics.com/Articles/Internet_of_Lights_Meets_Industrial_Internet_of/a56860)

- LiFi
 - “Li-Fi is a wireless communication technology which utilizes light to transmit data and position between devices.” (<https://en.wikipedia.org/wiki/Li-Fi>)



“INTERNET OF LIGHTS” - WORKSHOP CO-LOCATED WITH MOBISYS 2021

Overview

Menu

Date: 25 June 2021

Internet of Lights

Time: 15:00 - 18:30 CEST

Privacy & Cookies Policy



- Internet of sounds
 - “The Internet of Sounds (IoS) is an emerging research field positioned at the intersection of the Internet of Things and Sound and Music Computing domains.”

(<https://internetofsounds.net/about/>)

Introducing Our Newest Conference: The IEEE International Symposium on the Internet of Sounds

We are pleased to announce that the **Internet of Sounds Symposium (IS²)** is now part of the IEEE Communication Society's conference portfolio. Now in its 5th edition, this conference covers an emerging research field at the intersection of the Sound and Music Computing and the Internet of Things domains.

IEEE IS² will be hosted at the International Audio Laboratories in Erlangen, Germany, a joint institution of the Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) and Fraunhofer Institute for Integrated Circuits (IIS). The event will consist of presentations, keynotes, panels, poster presentations, demonstrations, tutorials, music performances, and installations.

Important Dates

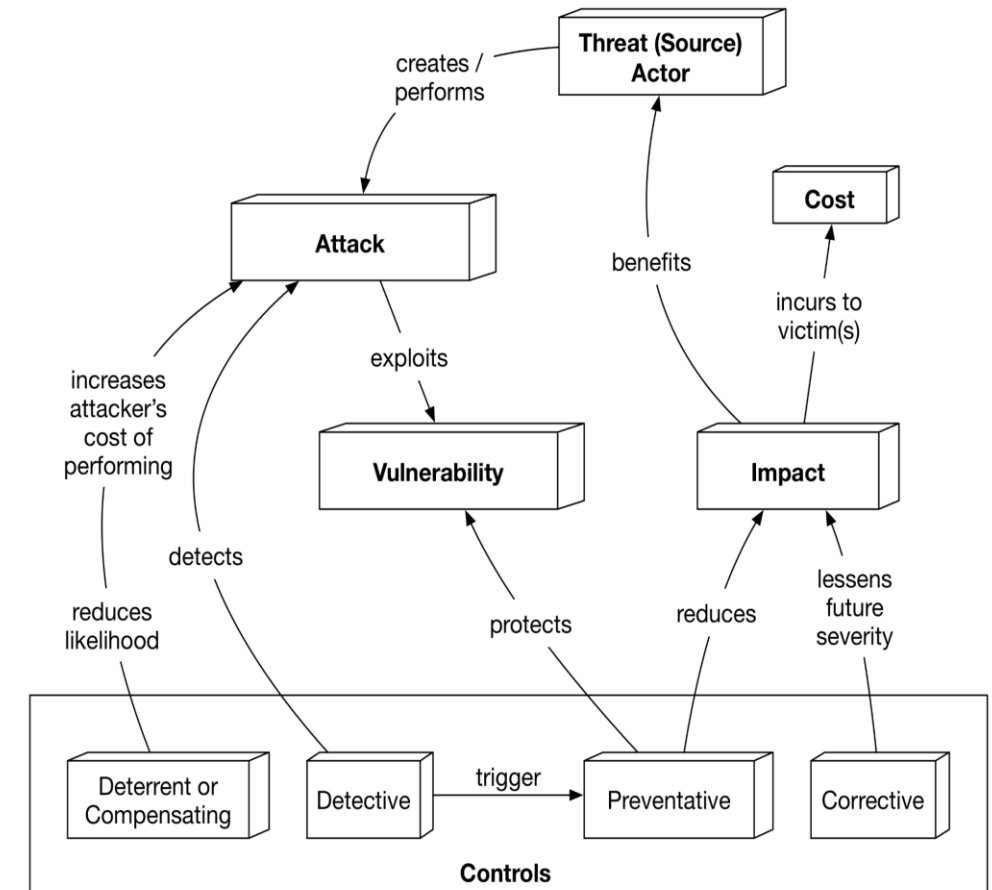
All Submissions Due:
10 July 2024

Author Notification:
15 August 2024

Camera-Ready Submission:
9 September 2024

Vulnerabilities, Attacks and Countermeasures

- Identify use cases (more, or less, innovative)
 - Application
 - Context
 - Framework
 - Environment
- Attacker model
- Realistic (and unrealistic) threats
- Intrinsic vulnerabilities
- Penetration testing
- Identification of attacks
- Implementation of countermeasures
 - Quality/performance of countermeasures
- Attack → countermeasure → new attack → new countermeasure → ...



Russel & Van Duren: "Practical Internet of Things Security", 2nd edition, Packt Publishing, 2019